Sajjad Chaus

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Experience

ML Engineer Intern — Dec – Feb 2026
Matrice AI — *Remote, India*

- Engineered core CV pipelines for customer analytics, optimized to reduce latency by **15%** for real-time streams.
- Fine-tuned custom YOLO models **(JointBDOE)** and accelerated inference via TensorRT for real-time streaming.
- Prototyped a Smart People Counter using **YOLOv11s**, achieving **0.92 mAP** on high-density mall datasets.

AI/ML Engineer (Internship) — May – Sept 2025
EngageOS.ai — *Remote, India*

- Led end-to-end development of an AI Agent-as-a-Service platform, enabling brands to automate multi-channel engagement and achieve 24/7 autonomous customer interactions.
- Architected scalable backend infrastructure using FastAPI + LangGraph, handling 1,000+ concurrent agent requests with 99.9% uptime and less than 3s median response time.
- Implemented advanced RAG pipelines with Qdrant VectorDB and Redis caching, reducing retrieval latency.
- Optimized AI decision-making workflows, integrating proactive + reactive agent logic, reducing operational cost.

Projects

MediTranslate – Healthcare Doctor-Patient Translation App | *FastAPI, React, Groq, WebSockets, PostgreSQL*

- Built a real-time voice and text translation platform supporting **20 languages**, enabling seamless doctor-patient communication with translated audio playback.
- Engineered a multimodal pipeline: Browser audio → Groq Whisper large-v3 (STT) → LLaMA 3.3 70B (medical translation) → Edge-TTS speech synthesis.
- Implemented role-aware translations preserving medical terminology and generated structured clinical summaries including symptoms, diagnoses, and follow-ups.

NovaML – Multi-Agent Data Science Platform | *LangChain, LangGraph, Python, Groq, FastAPI, Streamlit*

- Built a multi-agent AI platform using LangGraph with **6 specialized agents** to automate data ingestion, preprocessing, model training, and evaluation.
- Implemented conditional workflow routing with LLM-powered reasoning for intelligent model selection and hyperparameter tuning.
- Designed a Human-in-the-Loop (HITL) approval system for validating critical workflow stage through a Streamlit(UI).

Distributed System Monitoring AI Platform | *FastAPI, Python, Docker, PostgreSQL, WebSockets, Scikit-learn*

- Architected a distributed real-time monitoring system with Python agents and a central FastAPI backend to ingest and analyze OS metrics (CPU, memory, network) across nodes.
- Developed an AI anomaly detection engine using Isolation Forest, reducing false positive alerts by **30–40%** compared to static thresholding; containerized full stack with Docker Compose.

Technical Skills

Programming Languages: Python, Java, SQL
AI Frameworks & Libraries: PyTorch, TensorFlow, LangGraph, LangChain, Streamlit, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Keras, Plotly
Backend Technologies: FastAPI, Flask, Celery, REST APIs, WebSockets, Asynchronous Programming
Databases: PostgreSQL, MySQL, Redis **VectorDB**: FAISS, Qdrant
Developer Tools: Git, GitHub, Docker, VS Code, PyCharm, JupyterNotebook
Coursework: Machine Learning, Data Analysis, Generative AI, Operating Systems, Networking, NLP, Computer Vision

Education

Thakur College of Engineering and Technology — Mumbai, India
B.Tech in Computer Science (IoT), Minor in AI — CGPA: 8.00 — *July 2022 – May 2026*

Achievements

- Ranked 4th in ML Fidlatica Machine Learning Competition
- Secured 363th position in Amazon ML Hackathon out of 82000+ participants